AI-ASSISTED TESTING

Using GitHub Copilot and Other Tools to Accelerate QA

Jeffery Payne | Chief Executive Officer | jeff.payne@coveros.com

Jeffery Payne

Jeffery Payne is CEO and founder of Coveros, Inc., a company that helps organizations accelerate software delivery using agile methods. Prior to founding Coveros, he was the co-founder of application security company Cigital, where he served as CEO for 16 years.

Jeffery is a recognized software expert and popular keynote speaker at both business and technology conferences on a variety of software quality, security, DevOps, and agile topics. He has testified in front of congress on issues such as digital rights mgmt., software quality, and software research.

WHO WE ARE

Better Software. Faster.

Coveros equips organizations to modernize software delivery while optimizing business value.

Achieve exceptional results through our integrated suite of services including: hands-on consulting, learning and coaching, industry leading events, and collaborative communities.

Coveros is a GitHub Premier Partner and a Microsoft Azure Accelerate Partner

Our Services

Software Transformation

Services to help you modernize your software process by improving agility, continuous delivery, test automation, and application security.

Software Development

Services to support the planning, design, development, testing, and delivery of secure, reliable software applications.

Immersive Learning

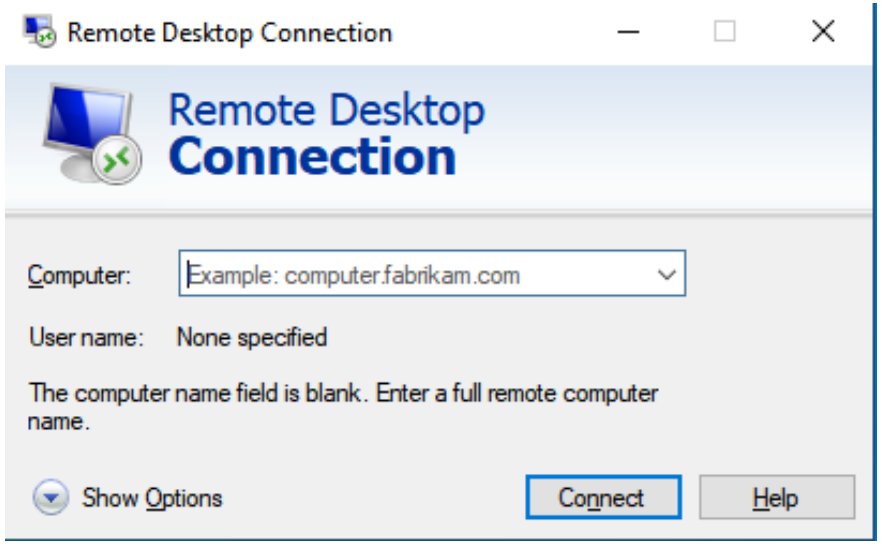
50+ hands-on training classes taught publicly and privately

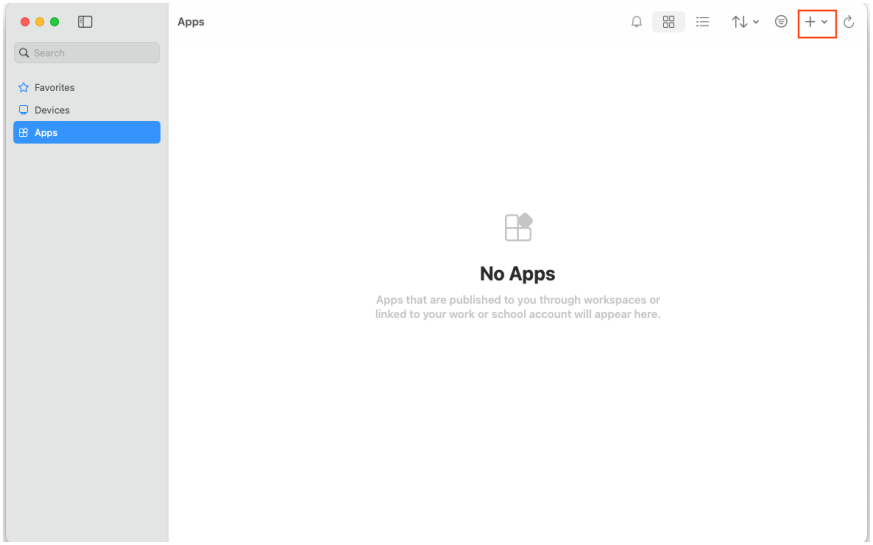
Enterprise, team, and individual coaching to reinforce concepts

Agenda

- What is AI-Assisted Testing?
- AI-Assisted Test Planning & Management
- AI-Assisted Test Case Generation
- Using Copilot Agents for Testing
- Wrap-up and Q&A

Test Environment Setup





1 What Is AI-Assisted Testing?

What is AI-Assisted Testing?

GitHub Copilot

Claude

codeium

Amazon CodeWhisperer

The process of leveraging Generative AI capabilities to assist in the testing of software.

CURSOR

codota

Continue

What Garter Says About AI-Assisted Software Development & Testing

63 percent of companies are in the process of testing, using, or have already implemented AI code assistants.

Gartner®

By 2028, 90 percent of enterprise software engineers will use AI code assistants.

Why Should Testers Care?

- **Testers can see the same type of productivity gains as developers by using AI**
- **Developers are potentially going to be creating more code faster, which means testing must accelerate too**
- **AI has the potential to help developers participate in the software testing process more than before**
- **AI is here to stay, so get used to it!**

The Good News

- **AI chatbots can help with a variety of testing tasks**
- **When properly trained, large language models (LLM's) can also generate good code (for both dev and test)**
- **The incorporation of AI into dev and testing tools provides a way to increase productivity**
- **Autocompletion of both test code and comments works seamlessly within interactive dev environments (IDEs)**

The Bad News

- **AI-generated application code and tests sometimes don't compile/run properly. AI is <u>NOT</u> a silver bullet**
- **The application code produced will sometimes have quality or security vulnerabilities**
- **Generated code may not comply with your dev and testing standards**
- **If the same model creates the application and the tests, will the tests effectively find bugs?**

Recent Published Research[1]

- **Productivity - 26% more tasks completed on average**
- **Code Commits - Weekly code commits increased by 13.5%**
- **Faster Iteration - Code compilation frequency rose 38.4%**
- **Quality - No negative impact on code quality was observed**
- **Junior Developer's - Less experienced developers saw the largest productivity gains**

[1]The Effects of Generative AI on High Skilled Work: Evidence from Three Field Experiments with Software Developers

Example Application

- A wine classifier application
- The application predicts varieties of wine based on their chemical properties
- The wine classifier leverages a trained AI machine learning model to make these predictions

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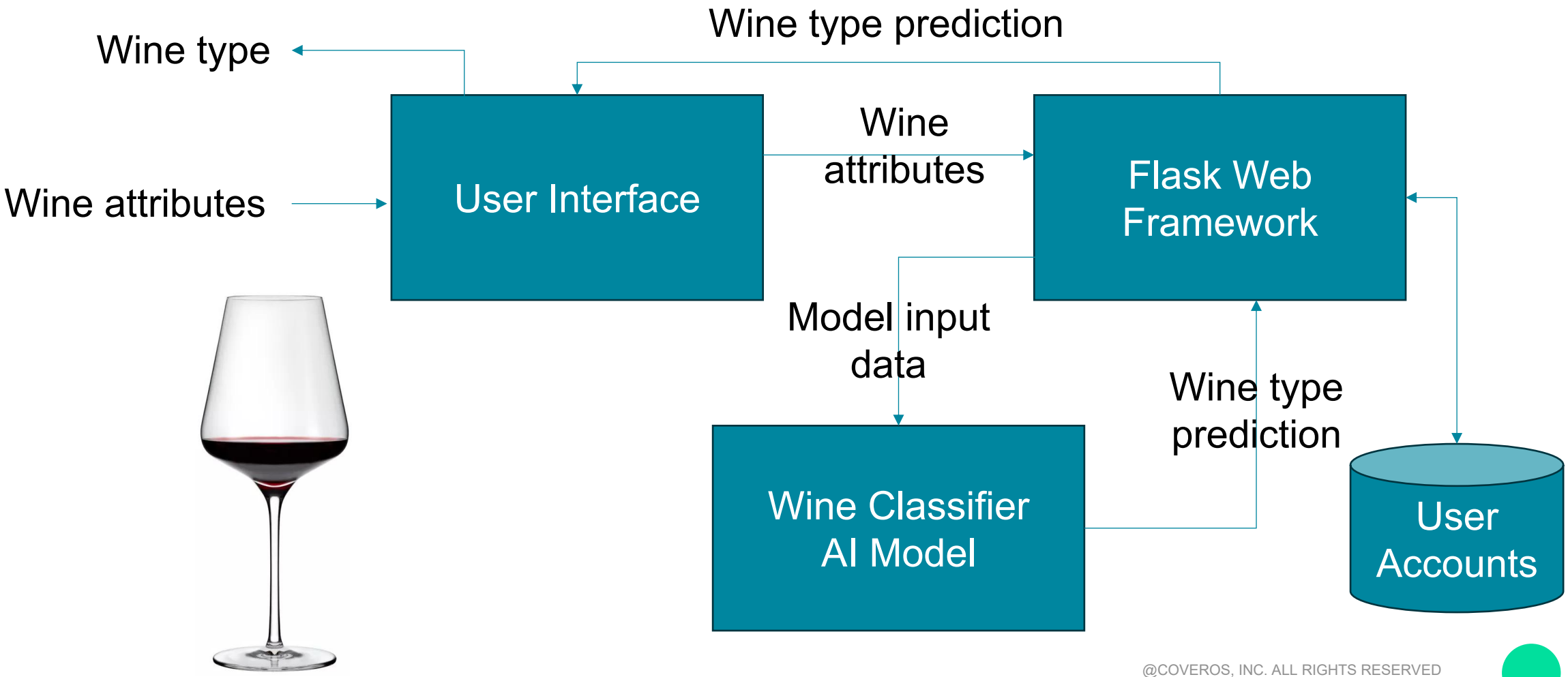
Product Architecture



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Intro to Visual Studio Code

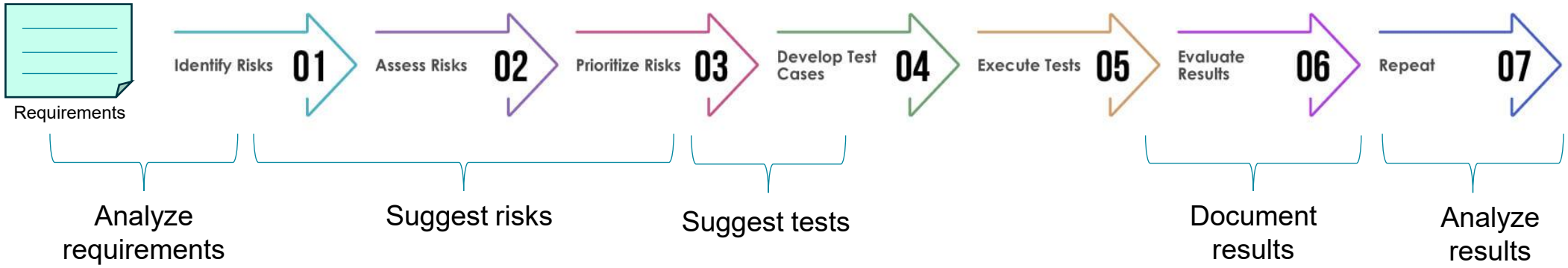
Visual Studio Code

- **A lightweight but powerful interactive development environment (IDE) that supports hundreds of programming languages**
- **Customizable with an extensive library of extensions**
- **Integrated debugging and version control**
- **Offers some amount of syntax highlighting, code completion, and automatic indentation out of the box**

2 AI-Assisted Test Planning and Management

AI-Assisted Testing Planning & Mgmt

Risk-based testing process



Where AI fits into test planning and management

AI-Assisted Testing Planning & Mgmt – Tool Support

Generative AI Chatbots

Gemini

ChatGPT

COPILOT

Intelligent Test Management Tools

spiraTeam

Katalon Studio

mabl

AI Frameworks for Agentic Solutions

Azure AI Studio

Microsoft Copilot Studio

Vertex AI

Amazon Bedrock

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Prompt Engineering – Best Practices for Software Testing

- **Principle #1: Write well-specified, testable prompts**
- **Principle #2: Encourage models to think**
- **Principle #3: Build good prompts iteratively**
- **Principle #4: Think beyond content creation**

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Using AI to Analyze Requirements

- **Do not try to analyze all your requirements at once!**

- **Key prompting principles that apply**
  - **Be focused and specific**
  - **Review existing work**
  - **Define your role**
  - **Use delimiters**

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Workshop Demo and Exercise

- **Follow along as the instructor shows how to identify and improve user stories**
- **Break into small groups**
- **Choose a user story and improve it using a chatbot**
- **Share and discuss your improvements with the class**

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Using AI to Suggest and Improve Risks

- Generate risks one at a time to focus your prompts
- Key prompting principles that apply
  - Be focused and specific
  - Define your role
  - Use delimiters to provide additional context
- Consider including information on the types of risks you are most concerned about in your prompt
- Give guidance on the structure of the results

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Using Models to Improve Prompts

- **Models represent some view of your system**
  - **Functional decomposition**
  - **Dataflow**
  - **Heuristic**
  - **Use case or workflow**
- **Understanding various representations for your system can help focus your prompts and get better results**
- **Use this to iterate on your risks and test cases during Sprints!**

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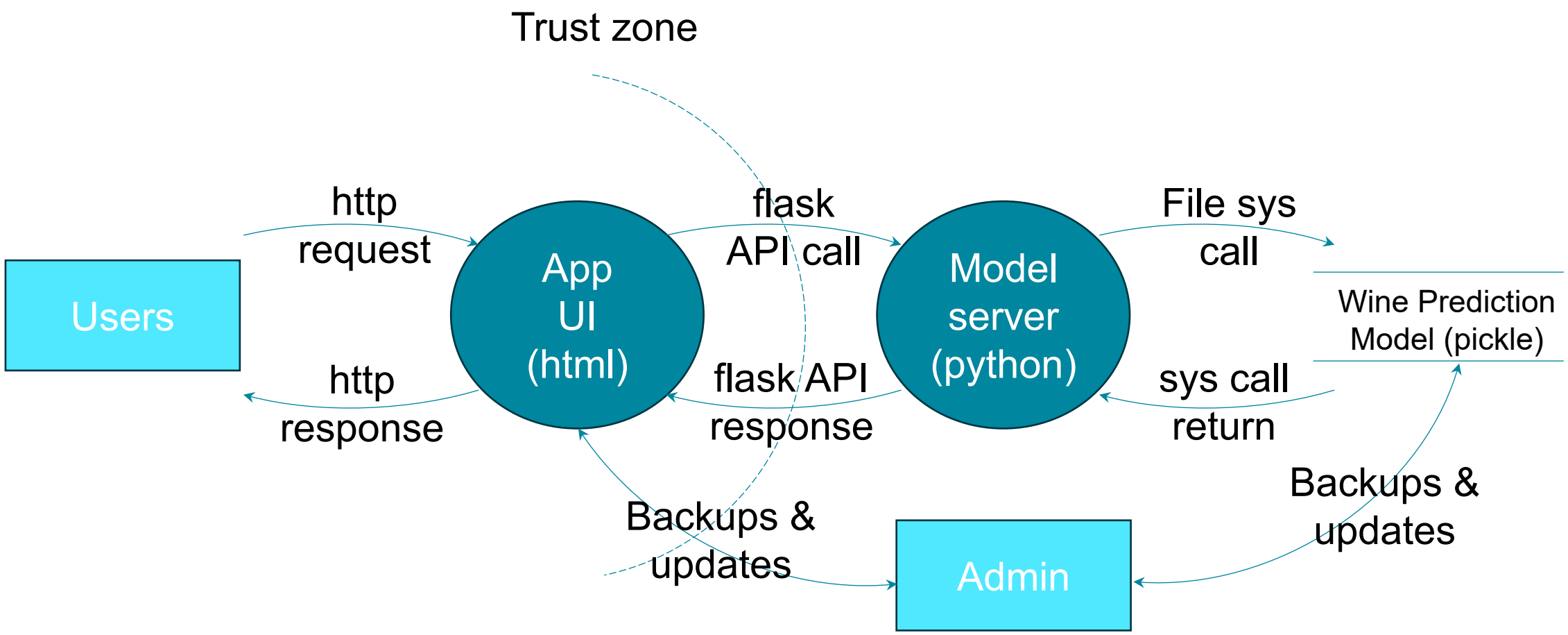
Using Models to Improve Prompts



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Workshop Demo and Exercise

- Follow along as the instructor generates and improves risks
- Break into small groups
- Reuse the user story you improved previously and generate some risks for it
- Share and discuss your improvements with the class

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Using AI to Generate Test Cases

- Tests are often split into several levels during test planning
  - Test requirements – the high-level description of a particular test
  - Test specifications – the detailed steps associated with a test requirement
- In agile, acceptance criteria for a user story are typically used to create test requirements
- Whether test specs are created is context-specific

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Using AI to Generate Test Cases

- **Consider leveraging the results of your requirements analysis, your risk analysis, AND any other related test cases you have created when prompting to create tests**
- **Send in context for tests you want to generate to improve results**
- **Consider leveraging intelligent test management solutions that will support this approach**

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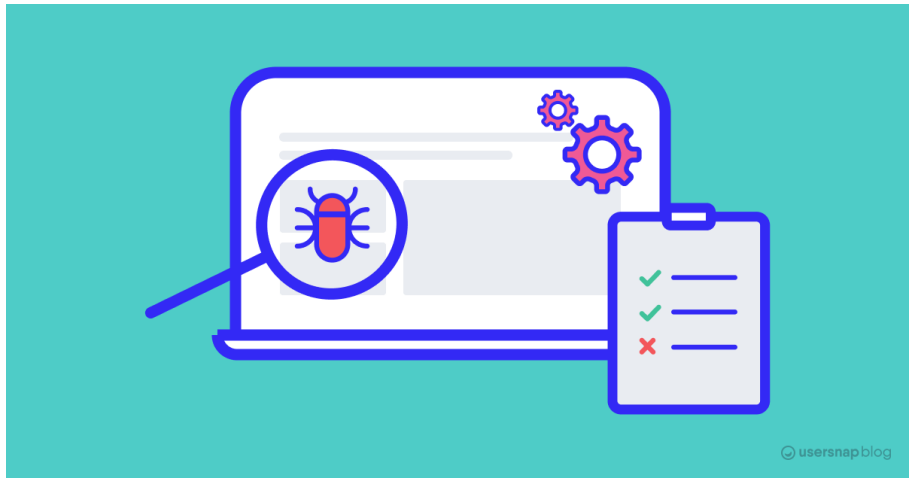
Workshop Demo and Exercise



- Follow along as the instructor shows how to generate tests
- Break into small groups
- Create tests for the user story you improved
- Share and discuss your improvements with the class

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3 AI-Assisted Test Case Generation

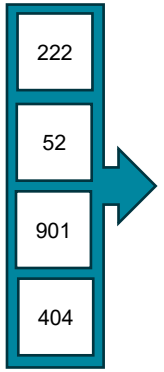
AI Support for Automated Test Case Generation

- Code-assistant for test code and comments completion
- Chatbot for automated test case generation
  - Prompts to generate automated tests
  - /tests shortcut
  - Refactoring of existing tests
- While generic LLMs can generate automated tests, models trained specifically on code are more accurate

How Code-Assistant Models Work

Input (existing code):

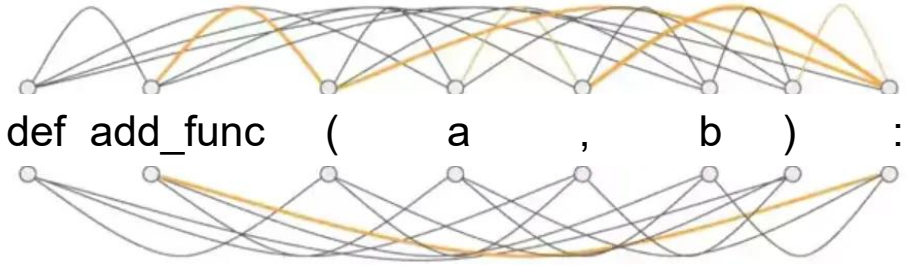
```
def subtract_func(a,b):
    c = a – b
    return c

def  add_func(a,b):
    |
```



Tokenizer

Generative Pre-trained Transformer (GPT)

Encoder (multi-head attention) → Decoder (multi-head attention)



New code

Prediction:

```
def subtract_func(a,b):
    c = a – b
    return c

def  add_func(a,b):
    c = a + b
    return c
```

AI-Assisted Code Completion

- **Aspects of Code Completion**
  - **Suggests completion of comments as you type**
  - **Suggests completion of line(s) as you type**
  - **Leverage comments to suggest code**

- **Most AI assistants use the Tab key to accept suggestions**

- **Many allow you to also choose a suggestion from a list of alternatives**

Introduction to GitHub Copilot

- Commercial code assistant integrated into common IDEs like Visual Studio Code
- Provides code completion, comment completion, chatbot capabilities to generate code, and much more
- Comes with its own coding model, or you can use an API Key to access your preferred model
- More accurate than using general chatbot models

Workshop Demo and Exercise

GitHub Copilot

- **Follow along as the instructor shows how to generate user story tests**
- **Break into small groups**
- **Create a user story test for one of the existing use stories**
- **Share and discuss your tests with the class**

AI-Assisted Test Case Generation

- There are a variety of ways to automatically generate test cases with AI-Assistants
  - Chat window
  - Chat line
  - Directed comments
- You can accept or reject test cases you generate
- Be specific in your prompts to get good results!
- Context and examples help a lot

AI-Assisted Test Refactoring

- **Refactoring is restructuring code to improve operation without altering functionality.**
- **Often done to clean up ‘code smells’ or adjust code to meet coding standards**
- **Most AI Assistants provide a mechanism for highlighting code and refactoring it automatically**
- **You are able to accept or reject suggested refactors**

What is SeleniumBase?

- **Built on top of Selenium Webdriver**
- **Simplifies web automation testing by offering pre-built functionalities such as:**
  - **Automatic waiting mechanisms**
  - **Page Object Model (POM) support**
  - **Easy-to-use assertions**
  - **Reporting and logging functionalities**
- **Works as a plugin for pytest for test integration**

AI-Assisted Test Case Generation - Exercise

pytest

SeleniumBase

- **Follow along as the instructor shows how to generate system tests**
- **Break into small groups**
- **Create a system test**
- **Use refactoring to improve it**
- **Share and discuss your tests with the class**

4 Using Copilot Agents

GitHub Copilot Agents

- **GitHub Copilot Agents is an advanced and integrated environment that goes beyond Copilot code completion and chat.**
- **It allows Copilot to directly interact with your GitHub repository to understand the context and propose more significant code changes.**
- **With Agents, Copilot is beginning to provide AI support for the entire SDLC**

GitHub Copilot Agents

- **Change GitHub Copilot to Agent model.**
- **Provide it any context beyond the folder and tabs you have open**
- **Describe a new test capability you want to add using Agents chat and follow the prompts**

GitHub Copilot Agents (cont.)

- **Agents will create a plan to do what you ask it to do that includes:**
  - **Identifying files that need to be modified or created.**
  - **Suggesting a high-level approach or architecture for the new feature.**
  - **Potentially breaking down the task into smaller, manageable steps.**

- **Automated code generation, including tests**

- **Propose changes as pull requests**

5 Wrap-up and Q&A

Continue Your AI Journey

Coveros AI Consulting Services

- AI Strategy Workshops
- AI System Implementation & Test
- Enterprise and Team AI Coaching
- Copilot Enablement & Adoption

Coveros AI Training Courses

- AI for Leaders
- Fundamentals of AI (certification class)
- AI for Testers
- AI and Machine Learning for DevOps

STAR WEST
A TECHWELL EVENT

QUESTIONS?

Thank you!

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